Course Code : ECT 359-2 **WYXZ/MS – 23 / 1604**

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination

INTRODUCTION TO MEMS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) Attempt all questions.

(2) Assume suitable data wherever needed.

(3) Support your answer with suitable diagrams if needed.

1. (A) Explain the basic components of MEMS. 6(CO1)

(B) Justify the requirement for an examination of the surface to volume ratio with reference to scaling rules in device geometry. 6(CO1)

2. (A) With relevant diagram write the working principle of operation of an electrostatic comb drive as a sensor and as an actuator. 6(CO2)

(B) Elaborate optical and electron beam lithography with reference to MEMS fabrication modules. 6(CO3)

3. (A) List the advantages and applications of RF MEMS resonators. 6(CO4)

(B) Describe the characteristics of etching, which removes material selectivity. 6(CO3)

4. (A) How do stress and strain relate to one another when a material exhibits elastic behavior ? 6(CO5)

(B) Write the advantages and disadvantages of finite element analysis. 6(CO5)

5. (A) Analyze the differences between thermal, chemical and biological sensors with respect to defence application. 6(CO2)

(B) Discuss the impact of Covid-19 Pandemic and Russia – Ukraine War on RF MEMS devices. 6(CO4)